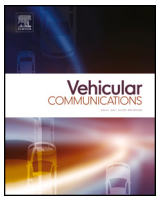
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A cooperative caching scheme utilizing regional feature and dynamic vehicle clustering in vehicular edge networks ☆

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ABSTRACT

Empowered with edge caching technology, the near-end storage resources of edge nodes in the Internet of Vehicles (IoV) can be fully utilized to accelerate the process of responding to content requests. In this paper, we propose a cooperative edge caching scheme utilizing regional feature and dynamic vehicle clustering (CRFDC). We take into consideration the fact that the change frequency of content popularity and regional feature is much lower than that of network topology and channel changes affected by vehicle mobility. To address this, we establish a double time-scale model. On the larger time-scale, we consider changes in content popularity and regional feature. On the smaller time-scale, we use the Prediction by Partial Matching (PPM) algorithm to predict vehicle's position. Additionally, we implement a dynamic cluster management approach, where vehicles with similar paths are grouped together, and use a consistent hashing algorithm to distribute contents among cooperative nodes. Finally, we employ deep reinforcement learning (DRL) approach to optimize our cooperative caching strategy for achieving lower content delivery latency. Simulation experiments demonstrate that our CRFDC scheme outperforms other cooperative caching schemes and benchmark algorithms in terms of reducing content transmission delay, improving cache hit ratio and decreasing communication overhead.

1. Introduction

With the widespread use of vehicle-connected devices and emerging vehicle-connected applications, it is predicted that the demand for mobile data traffic and bandwidth resources will explode exponentially which will lead to a series of problems, such as network congestion, high latency and poor user experience [1], [2]. In the traditional vehicle networking architecture, the requested content must be obtained from a distant Internet content center server before being distributed on the mobile core network. This not only results in long service delays but also places a heavy burden on backhaul links, potentially leading to congestion in the backbone network [3]. To address the demands of the intensive business requests in the Internet of Vehicles (IoV), more fine-grained content caching schemes are needed.

By implementing edge caching technology in vehicular networks, a well-designed cooperative caching scheme can be employed to fully utilize the storage resources of vehicles and RSUs. This allows users to access content from these nodes through vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communications. This method provides robust support for the various on-board service requirements of passing vehicles, including low latency, high bandwidth, and environment-awareness [4]. Research on vehicle networking cooperation cache can be divided into three directions: cooperation between vehicles [5], [6], [7], cooperation between vehicles and road side unit (RSU) [8], [9], [10] and cooperation between RSUs [11], [12], [13]. Huang et al. [14] studied a cooperative decision-making optimization method for content caching in IoV based on V2V cooperation to minimize the content acquisition delay for users. Wang et al. [15] employed content request prediction to optimize cooperative caching scheme, effectively handled dynamic and concurrent service requests in IoV. Su et al. [16] investigated the characteristics of content requests, including content access patterns, speed of vehicle and urban traffic density, and employed cross entropy to design a caching scheme in a dynamic content situations. However, none of the aforementioned works can consider the cooperation between different RSUs according to the urban traffic areas, and

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how to maintain stable node links when clustering vehicles in high-speed environments.

During the driving process, vehicles will pass through various areas with different road traffic conditions and traffic hub patterns. This can lead to variations in the social attributes and regional features of the vehicles [17]. In these areas, there are edge cache nodes available, such as RSUs and intelligent vehicles. However, effectively utilizing the vast and scattered cache space of these edge nodes is a challenging research topic in the field of edge computing for IoV. One of the main difficulties is designing a cooperative cache scheme that can make the most of the available resources. This requires dynamically determining the cache content and location, taking into account the time-varying characteristics of content popularity and validity. Additionally, previous research on the caching in IoV has focused on popular content in isolation, without considering cooperation between cache nodes. This results in a waste of potentially idle cache resources in adjacent nodes.

To deal with the above problems, considering time-varying content popularity, regional feature, dynamic network topology and vehicle clustering, we have investigated the optimal multi-node cooperative caching scheme in a vehicle-mounted edge caching network with limited storage capacity to enable the edge caching nodes to fulfill most content requests from vehicle users. The main contributions of this paper are listed as follows:

- We obtain regional feature by predicting traffic flow and calculating degree centrality, which are used to characterize the road traffic conditions and the importance of transportation hubs in a certain region. The RSU cooperation sets are divided based on regional feature, which has not been discussed in previous studies.
- We use the partial matching prediction algorithm to predict the next moment location of vehicles, perform dynamic clustering of vehicles, and assign content to RSU and vehicle cooperation nodes through consistent hashing.
- Based on the work described above, we have developed a double time-scale cooperative caching framework that utilizes deep reinforcement learning (DRL) to improve performance. By utilizing DDQN, the caching nodes are able to determine the most efficient cooperative caching scheme, resulting in reducing average delay for content transmission.

The remainder of this article is organized as follows. In Section 2, we review related work. In Section 3, we introduce the system model. In Section 4, we propose a cooperative caching scheme utilizing region feature and dynamic vehicle clustering. In Section 5, we present simulation results. Finally, in Section 6, we conclude the paper and discuss future work.

2. Related work

As a mobile edge service, edge caching has garnered significant attention in research due to its potential to alleviate the cache capacity limitation of mobile devices. We firstly provide an overview of progress made in edge caching schemes for both mobile and fixed terminals. We will then delve into the current state of research on edge cooperative caching schemes in IoV. Sun et al. [18] utilized user's spatial location and mobile information to construct user contact mobility model, and developed a mobile-aware D2D caching strategy for reducing average file transfer latency. In a similar vein, a cooperative caching strategy utilizing user's mobile trajectory, duration of stay and request intent was proposed to dynamically perceive the cache value and select the cache content [19]. In [20], considering the cache placement and external constraint parameters in D2D networks, the authors proposed a mobile-aware caching scheme aimed at minimizing user delay and maximizing system performance. In [21], the authors proposed a cooperative caching approach in content centric networks that optimize performances such as latency and cache hit ratio by cooperating among the routers in the neighborhood. Additionally, Zhou et al. [22] focused on joint optimization of edge caching and resource allocation in mobile edge computing (MEC) and proposed a Deep Deterministic Policy Gradient (DDPG) based algorithm to effectively reduce the system energy consumption.

The reasonable application of edge cache technology in IoV can provide strong support for a large number of vehicle service needs generated by passing vehicles. In [23], cooperative and non-cooperative paradigms were examined for vehicular demands and mobility patterns, using priori information and under both highway and urban traffic models. Xing et al. [24] utilized the Hawkes process to adjust to the constantly shifting popularity of content and optimize data throughput from edge devices. Lin et al. [25] proposed a greedy cooperative caching response algorithm that adjust the allocation order of caching nodes the constraint of node topology. But [23], [24], [25] only considered the correlation between content popularity and time, or assumed that the content popularity is given. As improved, He et al. [26] proposed a spatial-temporal correlation algorithm to improve the accuracy of content popularity prediction by analyzing the spatio-temporal correlation of historical content requests. Further more, addressing the issue of reduced caching efficiency caused by cache redundancy among edge nodes, Wu et al. [27] proposed a hybrid content value cooperative caching scheme leveraging temporal convolutional network, which realizes the mutual cooperation between different edge nodes.

V2V communication facilitates fast connection and low communication latency between communicating vehicles, which provides convenience and support for users to quickly obtain required content and realize edge cooperative caching scenarios of IoV [28]. Moreover, Vehicles in IoV may have different traffic attributes, including speed, traffic density, and travel time in same direction. These attributes are crucial factors in the edge cooperative caching system, which significantly impact the communication distance, channel quality and data transmission duration of the target vehicle [29], [30], [31]. Huang et al. [32] proposed a cooperative caching approach by vehicle clustering to build communication within and between vehicle clusters, which can effectively improve cache performance. Sou et al. [33] proposed a content distribution strategy with clustering algorithms, which considered the cooperation between RSUs in different geographic locations to improve the variety of cluster cache space content. Due to the constantly changing nature of networks and complex environment of IoV, it is crucial to accurately optimize cache placement and delivery in order to account for network conditions and storage limitations. Thereinto, deep reinforcement learning (DRL) has been used as an effective technology for content caching in IoV [34], [35], [36], [37], [38]. In [39], exploring the cooperation between vehicles and RSUs on content scheduling, a mobile edge computing and caching scheme with social awareness was proposed to improve content processing and caching. Qiao et al. [40] proposed a cooperative caching strategy utilizing the deep deterministic policy gradient framework considering that the update speed of content timeliness is much less than that of vehicle driving environment and network topology.

Although the above researches provide some insights into edge services and cooperative caching for IoV, no research has been conducted on the edge network as a whole. The edge caching schemes based on vehicle caching have not taken into account the very short connection time between vehicles in vehicle clustering. Traditional vehicle clustering is difficult to maintain link stability, which may cause data transmission failure. The edge caching schemes utilizing fixed infrastructure such as RSU have not considered the impact of road traffic conditions in the region where the edge infrastructure is located on vehicle requests and cooperative caching systems. In order to solve the above challenges, we proposes an innovative edge cooperative caching scheme that takes into account regional feature and dynamic vehicle clustering.

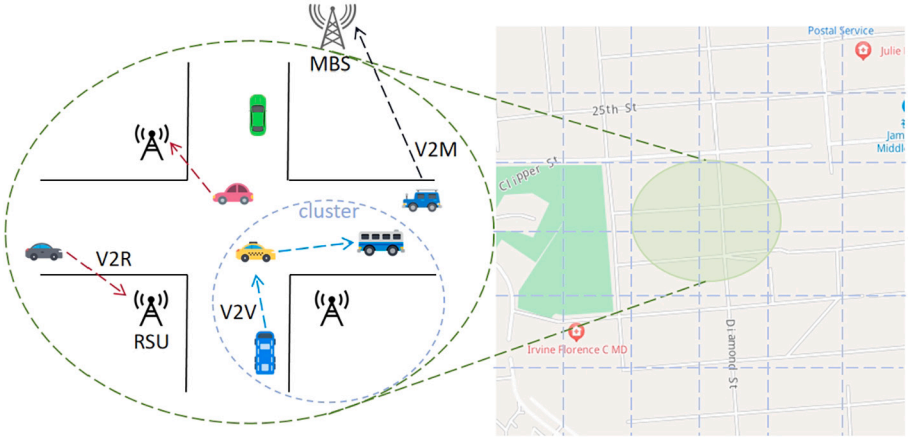


Fig. 1. Edge cooperation cache network scenario.

Table 1
Major Notations.

Notations	Definition
V_k	Content requesting vehicle
V_j	Content caching vehicle
R_l	Local RSU
R_c	Cooperative RSU
C_f	The content indexed to f
ρ_{C_f}	Popularity of content C_f
I_{C_f}	The number of times C_f is requested
$T_{k,j}$	Daley for V_k obtain requested content from V_j
$T_{k,l}$	Daley for V_k obtain requested content from R_l
$T_{k,c}$	Daley for V_k obtain requested content from R_c
$T_{k,M}$	Daley for V_k obtain requested content from MBS
G_n	The n-th region
d_{G_i}	Traffic flow of region G_i
D_{G_i}	Degree centrality of region G_i
$\rho(G_i)$	Region feature of region G_i
B_k	Bridging centrality of V_k
c_k	The number of vehicles within the V2V communication range with V_k

3. System model

3.1. Network scenario

Fig. 1 shows the edge cooperative caching network scenario of IoV. We have chosen discrete geographical regions instead of infinite continuous locations to facilitate obtaining results including vehicle movement patterns and regional feature. In our architecture, there are different types of edge caching nodes, consisting of macro base station (MBS), RSUs and intelligent vehicles, each RSU or vehicle is equipped with wireless communication capability and limited storage capacity.

As vehicles travel, they pass through multiple regions, generating a variety of data such as road congestion conditions, vehicle driving patterns, and vehicle social attributes. This results in varying road traffic conditions and levels of traffic congestion between each region. RSUs will record these data and ultimately obtain regional feature, which can then be used to determine the cooperation set. Let $R = \{R_1, R_2, ..., R_l\}$ represent the index set of RSUs, $V = \{V_1, V_2, ..., V_k\}$ represent the index set of vehicles and $C = \{C_1, C_2, ..., C_f\}$ is the index set of contents. We consider that the vehicle communicates with caching node via wireless link, and RSUs are connected to each other through wired links. For ease of reading, the major notations used in this paper are listed in Table 1.

Assuming that MBS has sufficient storage capacity to cache all available content, RSU has limited storage capacity to accommodate only part of the content, and intelligent vehicles have the least storage capacity. The vehicle can obtain the requested content file from nearby intelligent vehicles, local RSU or cooperative RSU, and MBS. The specific content distribution process is as follows:

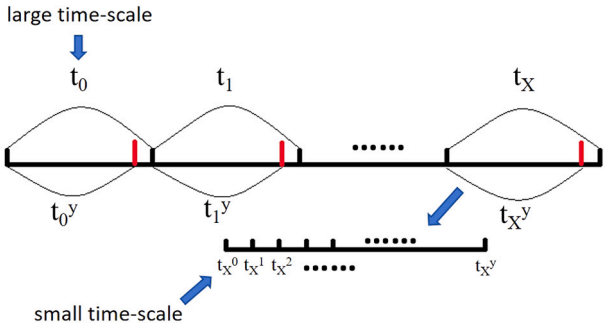


Fig. 2. Double time-scale caching model.

1. Intelligent vehicle node: When requesting content, the vehicle will first search for nearby intelligent vehicles that are capable of V2V communication. If a node has cached the content file, it will be transmitted to the vehicle through V2V. Despite the limited storage capacity of intelligent vehicles, the close proximity of vehicles during V2V communication minimizes content distribution delays through the intelligent vehicle node.
2. RSU: Vehicle will attempt to obtain file from RSU when all of the available V2V nodes do not cache the required content. The RSUs determine the cooperation set according to the value of regional feature. If the requested file is already cached in the local RSU, the local RSU will send the content to the requesting vehicle through V2R. If only the cooperative RSU caches the content, the local RSU would fetch the file from the cooperative RSU with wired connect and forward it to the vehicle. Compared with intelligent vehicles, RSUs have larger cache capacity and can distribute content to all vehicles within coverage with lower latency.
3. MBS: If the vehicle is unable to receive the requested content through V2V and V2R modes, it will send a request to the MBS. The MBS will then transmit the content file to the vehicle through V2M. The MBS stores all available content and can fulfill requests from all vehicles. However, due to its distance from the vehicle, the MBS has the highest latency. Therefore, it is important to minimize the use of the MBS for content distribution.

3.2. Double time-scale caching model

Since the change frequency of content popularity and regional feature is much slower than that of network topology and channel changes affected by vehicle mobility, we models cooperative caching as a dual-time-scale content update and distribution process. As shown in Fig. 2, the content update and distribution processes occurs at different time granularities. On the larger time scale t_x, $\{t_x | x \in \{0, 1, ..., X\}\}$, content

popularity, road traffic conditions, and traffic hub patterns remain relatively stable. At the beginning of each t_x period, the MBS updates the global content popularity, and selects the RSU cooperation set based on the vehicle driving conditions in each region during the previous period, t_{x-1}, to make content placement decisions for the RSUs. On the smaller time scale $t_x^y, \{t_x^y | x \in \{0,1,...,X\}, y \in \{0,1,...,Y\}\}$, while distributing content to requesting vehicles, vehicle route is calculated and predicted and vehicles with analogous route over a period of time will be clustered together.

Most researches assume that content popularity follows the Zipf law in daily social networks and remains static [41], which is not suitable for IoV with high mobility and frequent region switching. Some studies adopt the time-varying content popularity, but the way to calculate content popularity is completely based on the content request rate within a certain period of time, which is not in accordance with the basic relationship law of social networks and cannot well reflect the content popularity distribution. Therefore, we considers the global content popularity composed of Zipf distribution and content requesting rate [42], which can be formulated as

$$\rho_{C_f} = \frac{f^{-\alpha}}{\sum_{m=1}^{F} m^{-\alpha}}, \tag{1}$$

where α is the exponent characterizing the Zipf law distribution and f is the sorted index of the number for C_f. We assume that content request interval follows the Poisson distribution. At the start of the larger time scale, MBS will track the number of requests for all content and update the popularity of the content. This can be formulated as

$$\rho_{C_f} = \mu \frac{I_{C_f}(t_x)}{M} + (1-\mu)\rho_{C_f}, \tag{2}$$

where μ is the time average constant, representing the weight reduction of previous polularity. $I_{C_f}(t_x)$ is the number of times C_f is requested at period t_x, and M is the total number of content requests.

3.3. Communication model

The demand for ultra-reliable and low-latency communications is particularly critical in the IoV scenario due to the high mobility of vehicles, frequently changing content requirements and harsh communication environment [43]. For the convenience of analysis, we assume that the network topology would stay the same in the file distribution time slot, and vehicles will use the orthogonal spectrum resources allocated by MBS to communicate with other nodes so that there is no interference among V2V, V2R and V2M communications.

In the network scenario we constructed above, the vehicle can obtain the requested content in three different ways, and obtaining the content from different content providers incurs different latencies, so we separately model the delay of receiving requested content from different nodes.

a. When requesting vehicle V_k receives content from caching vehicle V_j in V2V mode, the signal-to-noise ratio (SNR) between V_k and V_j is given by

$$SNR_{k,j} = p_j G_{k,j} \sigma_c^{-2}, \tag{3}$$

where p_j represents the transmit power of V_j. $G_{k,j} = \alpha_{k,j} d_{k,j}^{-\zeta}$ is the channel gain between V_k and V_j, $\alpha_{k,j}$ represents the path loss constant, $d_{k,j}$ and ζ represent the distance of V2V communication and path loss index, respectively, σ_c^{-2} is the reciprocal of the noise power. Thus, the delay for V_k to obtain the requested content from V_j can be calculated as

$$T_{k,j} = \frac{F}{w_{k,j} log_2(1 + SNR_{k,j})}, \tag{4}$$

where F represents the content file size, $w_{k,j}$ represents the available bandwidth between V_k and V_j.

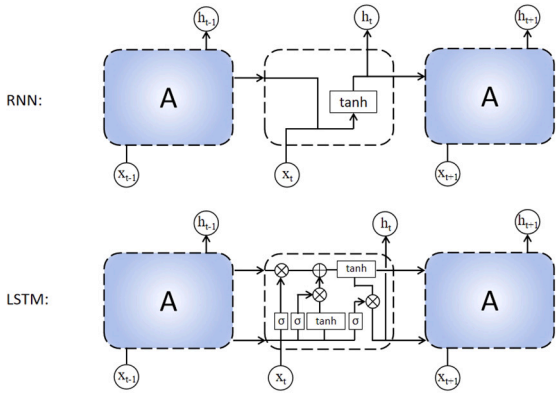


Fig. 3. Structure of RNN and LSTM.

b. If V_k cannot get the content in V2V communication, the vehicle will try to get it from local RSU (R_l), the SNR between V_k and R_l is given by

$$SNR_{k,l} = p_l G_{k,l} \sigma_c^{-2}, \tag{5}$$

where p_l represents the transmit power of R_l, $G_{k,l} = \alpha_{k,l} d_{k,l}^{-\zeta}$ is the channel gain between V_k and R_l. The delay for V_k to obtain content from R_l can be calculated as

$$T_{k,l} = \frac{F}{w_{k,l} log_2(1 + SNR_{k,l})}, \tag{6}$$

where $w_{k,l}$ represents the available bandwidth between V_k and R_l. If only the cooperative RSU (R_c) caches the requested content, R_c will first transmit file to R_l through wired connection, which then delivers it to V_k. Similarly, the delay can be calculated as

$$T_{k,c} = \frac{F}{r_{l,c}} + \frac{F}{w_{k,l} log_2(1 + SNR_{k,l})}, \tag{7}$$

where $r_{l,c}$ represents the transmission rate between R_l and R_c.

c. Moreover, if V_k does not receive the requested content in both V2V and V2R communication, the MBS will transmit it to V_k through V2M, the SNR between V_k and MBS is given by

$$SNR_{k,M} = p_M G_{k,M} \sigma_c^{-2}, \tag{8}$$

where p_M represents the transmit power of MBS, $G_{k,M} = \alpha_{k,M} d_{k,M}^{-\zeta}$ is the channel gain between V_k and MBS. The delay for V_k to receive content in V2M mode can be calculated as

$$T_{k,M} = \frac{F}{w_{k,M} log_2(1 + SNR_{k,M})}, \tag{9}$$

where $w_{k,M}$ represents the channel bandwidth between V_k and MBS.

4. Cooperative caching scheme

In this section, our CRFDC scheme will be introduced in detail, including defining regional feature, dividing vehicle clusters, explaining the method of content allocation, and discussing the use of DRL to optimize cooperative cache scheme.

4.1. Regional feature

Due to the high mobility, intelligent vehicles will pass through multiple areas while driving on the road, resulting in the varying traffic information produced in different areas [44]. We have respectively chosen traffic flow and degree centrality to accurately represent the road traffic conditions and the importance of transportation hubs, which has not been mentioned in the previous IoV researches.

1) Traffic Flow Forecast: Vehicles can only drive on specific roads because of the restriction on road network conditions which are interlaced

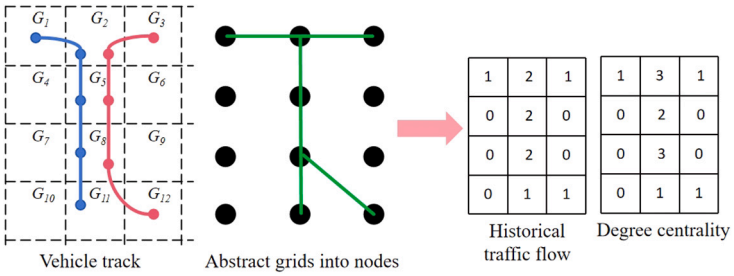


Fig. 4. Traffic flow and degree centrality.

and complex. The road traffic conditions in different regions will affect each other and lead to its spatial correlation. For a given area, the traffic conditions in the previous period will have an impact on the next period, resulting in a time correlation. To account for both temporal and spatial correlations in road traffic conditions, we utilize machine learning techniques to uncover hidden patterns in data based on the historical traffic flow of each region. This allows us to establish a prediction model that can accurately forecast future traffic flow. However, traditional Recurrent Neural Networks (RNNs) are not only struggling to capture long-term dependencies but also face problems such as vanishing gradient and exploding gradient. In consideration of the above concerns, we utilize Long Short-Term Memory (LSTM) to predict the traffic flow.

As illustrated in Fig. 3, LSTM addresses the issue of long-term dependencies problems in traditional RNNs by incorporating a gating mechanism and cell state. This allows for the transfer of long-distance information within the network, making it well-suited for processing time series data. In summary, LSTM can effectively predict the traffic flow of each region in the next time period. The input for the time series is the normalized traffic flow of each region in the previous $x-1$ time periods, represented as $d = \{d(t_0), d(t_1), ..., d(t_{x-1})\}$. The output is a prediction sequence for the next time period, expressed as $d^*(t_x)$. To obtain the predicted traffic flow for each region in the next larger time scale, $d(t_x)$ can be reverse-normalized.

2) Degree Centrality: Because of the uneven distribution of roads, some areas with pivotal intersections are obviously more momentous in terms of traffic, different areas will have different degrees of transportation hub condition. While the road traffic conditions and the importance of transportation hubs are closely related, they are not identical. As shown in Fig. 4, degree centrality is defined as a extent to which a node is connected to all other nodes. The higher the degree centrality, the greater the influence of the node in the network. In this article, degree centrality is determined based on the number of vehicle trails passing through region G_i during period t_{x-1}. This can be expressed as

$$D_{G_i}(t_x) = \sum_{j=1}^{N} x_{ij},\ i \neq j, \tag{10}$$

where x_{ij} represents the edge connecting other nodes are connected to node G_i.

The significance of degree centrality in determining the importance of a region becomes more apparent when there are fewer vehicles present. In Fig. 4, the traffic flows of grids $\{G_2, G_5, G_8\}$ are equal when there are only two vehicle trails, making it difficult to distinguish the differences between these grids. However, from the perspective of degree centrality, it is evident that G_2 and G_8 have a higher degree than G_5. This indicates that G_2 and G_8 are more crucial at the transportation hub level, as they experience more pressure on network traffic.

3) Region Feature: We define regional feature composed of traffic flow and degree centrality to determine the RSU cooperation set, which can characterize the road traffic conditions and the importance of transportation hubs in the region. For region G_i, its regional feature can be calculated as

$$\rho(G_i) = \alpha \frac{d_{G_i} - \overline{d_{G_i}}}{std(d_{G_i})} + \beta \frac{D_{G_i} - \overline{D_{G_i}}}{std(D_{G_i})}, \tag{11}$$

where α and β represent the weights of traffic flow and degree centrality respectively, with the condition that $\alpha + \beta = 1$.

A grid with a higher regional feature means a greater network traffic burden, which can be effectively alleviated by implementing edge cooperative caching. The RSU with the lowest regional feature among its neighboring RSUs will be selected as the cooperative set for RSUs with higher regional feature, which can store more content files, reducing cache redundancy. Each RSU can have up to four cooperative neighboring RSUs at a time. The selected RSU will not participate in the cooperation set of other RSUs. However, as vehicle density increases and the regional features of adjacent grids become closer, the performance improvement from cooperative caching may not be enough to compensate for the additional delay caused by dividing RSUs into cooperative sets. Therefore, the selection of cooperative RSUs must be made in real time based on the distribution of road traffic conditions.

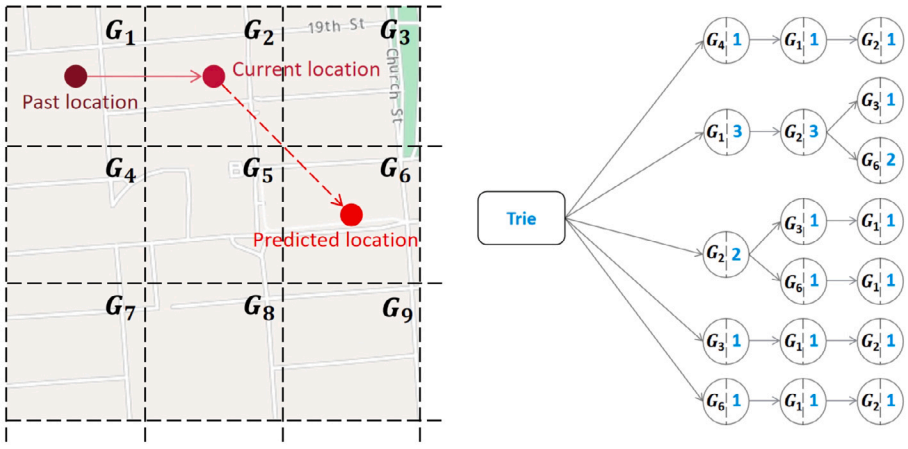
4.2. Dynamic vehicle clustering

To avoid the additional signal overhead and interference caused by V2V broadcast communication, we utilized dynamic vehicle clustering on the small time-scale by predicting the position of all vehicles in the next time slot, while improving the communication stability between vehicles.

As illustrated in Fig. 5(a), we used discrete geographical regions instead of infinite continuous locations to facilitate calculating vehicle positions and regional features. By discretizing the coordinates of each vehicle and convert them into corresponding grids, we can obtain a set of grids representing the route of vehicle over a period of time, $S = \{G_1, G_2, ..., G_n\}$, where G_n represents the nth grid. Then, we construct a tree structure trajectory model based on S as shown in Fig. 5(b), and utilize the prediction by partial matching (PPM) algorithm to forecast the position of vehicle at the next time period t^{y+1}.

In the tree structure model, the historical driving trajectory is converted into grid sequences as input, thus obtaining a root node and multiple child nodes, each of which consists of two elements: the grid position G_n and a counter that records the number of times G_n appears in S. Each branch of the tree is a trajectory sequence consisting of three consecutive grid positions. In the input process of tree model, if a certain trajectory sequence already exists as a branch, the counter of each child node in that branch is incremented by 1 instead of adding a new branch. For instance, if the given trajectory is $\{G_4, G_1, G_2, G_3, G_1, G_2, G_6, G_1, G_2, G_6\}$, the input grid sequences for model would be $\{G_4, G_1, G_2\}, \{G_1, G_2, G_3\}, ..., \{G_1, G_2, G_6\}$. The sequence $\{G_1, G_2, G_6\}$ will be only recorded once in the tree even though it appears twice in grid sequences.

When predicting vehicle trails, it is necessary to determine the position of the vehicle at time slots t^{y-1} and t^y. In order to obtain the most likely region $G(t^{y+1})$ for the next time period, the model should be able to resolve the sequence of locations $G(t^{y-1})$ and $G(t^y)$. For instance, if the given sequence $\{G(t^{y-1}), G(t^y)\}$ is $\{G_1, G_2\}$, according to the second branch in the tree constructed in Fig. 5(b), it can be calculated that



(a) Prediction by partial matching (b) Trajectory tree model

Fig. 5. Dynamic vehicle clustering.

the vehicle has a 1/3 probability of arriving at G_3 at t^{y+1}, and a 2/3 probability of arriving at G_6, thus the predicted position would be G_6. If there are multiple positions with the same probability, the RSU will select the most likely region based on the historical driving trail of other vehicles.

In dynamic vehicle clustering, it is vital to choose a stable center node maintaining the cluster. To measure the stability of a vehicle's connection with other vehicles, we use the bridging centrality as an indicator, which is defined as the degree of control a node has over the flow of information between adjacent nodes. Nodes with high bridging centrality have stronger connectivity and stability [42]. In this paper, the bridging centrality of V_k can be expressed as follows:

$$B_k(t_x^y) = c_k \cdot \frac{N_k}{N}, \tag{12}$$

where c_k represents the number of vehicles within the V2V communication range with V_k at time period t_x^y. N represents the trail sequence of all vehicles within the V2V communication range. N_k represents all trail sequences in N that will pass through V_k at the next time slot, $N_k \subseteq N$. Each RSU selects one vehicle with the largest bridge centrality within its coverage range as the cluster center. In cases where there are multiple vehicles with the same maximum bridge centrality, the RSU will calculate the average physical distance from each vehicle to other adjacent vehicles. The vehicle with the smallest average distance will be selected as cluster center. In order to maintain stability between the cache content and the cache nodes, and to avoid significant signaling overhead and interference caused by vehicles traversing all surrounding V2V communication vehicles in search of requested content, each RSU will select half of the vehicles in each cluster as cooperative cache nodes based on bridging centrality, since nodes with higher bridging centrality can maintain the stable connections with other nodes.

4.3. Content distribution

To avoid cache duplication between nodes in cooperative caching, we uses consistent hashing algorithm to distribute content to cache nodes. This algorithm maps fixed requests to fixed servers for processing by using a modulo operation on the total number of nodes. This approach enables the mapping of the relationship between service requests and service processors, achieving dynamic allocation. However, the inconsistency of this method renders it incapable of load balancing the entire system. In our CRFDC, the size of a cooperative cluster is not constant but rather undergoes changes over time. This is due to members leaving or joining, resulting in a change in the total number of nodes. Additionally, expansion or shrinkage will cause all previous hash mappings to be recalculated since data is stored on each node. As a result,

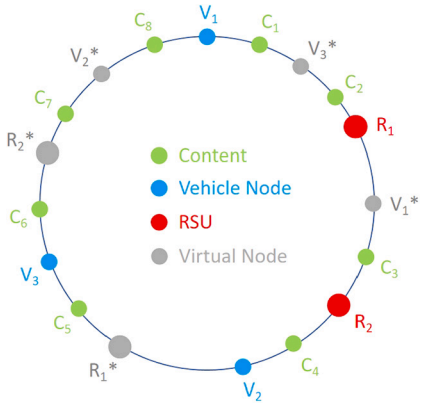


Fig. 6. Structure of hash ring.

even a small change in the cluster size can lead to a large amount of data migration, requiring the readjustment of all the data in the cluster. As the cluster size grows, the migration cost will also increase linearly. The object of the modulo operation in consistent hashing is a fixed value of 2^{32-1}, eliminating the possibility of rehashing issues. Compared to basic hashing, consistent hashing offers more balanced data distribution and stronger scalability, resulting in better load balancing and higher fault tolerance rate [45].

The hash ring is a virtual ring that is mapped by a consistent hashing algorithm. Each cooperative node serves as a placeholder for one or more requests. Unlike traditional hashing methods, the number of cooperative nodes is not fixed and the ring has an infinite number of points. Nodes can be placed at any position on the ring, and a hash function is used to convert the node's information into a hash value, which is then placed on the hash ring. Adding or removing a node will only affect its clockwise neighboring successor nodes on the ring, ensuring that no other data is affected. Content on the hash ring is distributed to the nearest node in a clockwise direction. However, when there are fewer cooperative nodes, consistent hashing can lead to an uneven distribution of cached content on a particular node. This issue can be addressed by adding virtual nodes to the hash ring [46].

The cooperation set of RSUs can be obtained on a large time-scale, taking into account regional feature. Meanwhile vehicle clusters can be obtained on a smaller time-scale through dynamic vehicle clustering. These cooperative RSUs, along with all the vehicle clusters within their respective coverage areas, collectively form a cooperative set. As shown in Fig. 6, there are 10 nodes on the hash ring, including two RSU nodes (R_1 and R_2) and three vehicle nodes (V_1, V_2 and V_3), as well as their corresponding five virtual nodes ($R_1^*, R_2^*, V_1^*, V_2^*, V_3^*$). Contents C_1 to

C_8 are mapped to different locations on the hash ring. Using consistent hashing, C_1 is distributed clockwise to the nearest virtual node V_3^*. Since V_3^* is a virtual node, C_1 will then be distributed to the actual node V_3. If a vehicle leaves the cluster, the content originally associated with that node will be distributed in a clockwise to the next node. The edge node will determine whether to cache the file upon its arrival, taking into consideration the content's popularity and the remaining cache space, without affecting the cached content of other nodes.

4.4. Optimal caching scheme based on DRL

The objective of this section is to optimize the cooperative caching scheme using DRL in order to minimize the average delivery delay of content. The selection of the cooperative RSU will impact the content allocation of the participating RSUs and all vehicle clusters within the coverage, ultimately affecting the system performance. It is important that the number of cooperative RSUs should not be maximized, as this may not necessarily improve the system's performance. Instead, the selection of cooperative RSUs should be determined in real time based on the distribution of road traffic conditions. Additionally, the content of the edge node cache should be dynamically updated and replaced. To address these issues, we propose using Double Deep Q Network (DDQN) to develop a DRL-based cooperative caching scheme. The DRL framework is outlined below:

(1) State: At the beginning of each large time-scale period, MBS will redetermine the RSU cooperation set according to regional feature, and then redistribute content based on consistent hashing. The movement of vehicles between regions results in a dynamic change of vehicle clusters, which in turn affects the cached contents of vehicle cooperative nodes. Accordingly, the state space can be defined as follows:

- $S_R(t_x^y)$: The total cached contents of RSUs at time slot t_x^y. RSUs are divided into different cooperative clusters based on regional feature, and their cached contents are sorted in descending order of content popularity.
- $S_V(t_x^y)$: The total cached contents of vehicle. The Vehicle nodes and their cached contents are ranked in descending order of bridging centrality and popularity respectively.

As a result, the joint system state can be expressed as

$$S(t_x^y) = \left\{[S_R(t_x^y)], [S_V(t_x^y)]\right\}, \tag{13}$$

(2) Action: The number of RSU cooperation is determined in real time according to the distribution of road traffic conditions. The cache node distributes content for vehicle requesting content at each content fetching time slot, and MBS determines whether the cache node needs to update the content according to the content popularity. The action space includes

- $\delta(t_x^y), \delta(t_x^y) \in [0, 4]$: The number of RSU cooperation at period t_x.
- $\sigma(t_x^y), \sigma(t_x^y) \in [0, 1]$: Whether to update the cache contents at time slot t_x^y. When the replacement action is performed, the n least popular contents will be replaced with randomly selected contents from the collection of hot content.

The joint action space is represented as

$$a(t_x^y) = \left\{[\delta(t_x)], [\sigma(t_x^y)]\right\}, \tag{14}$$

(3) Reward: The reward is generated based on the action in a certain state and utilized to adjust the behavior policy of agent in the training process. In this paper, the reward consists of the content transmission delay. The vehicle can obtain the requested content from caching vehicle, RSU and MBS in three ways: V2V,V2R, and V2M, respectively. Although the MBS has the capacity to provide all available content to vehicles, it is the most distant from the vehicle and takes the longest to transmit the same content. Therefore, content transmission via V2M should be avoided as much as possible. In this paper, the reward function is formulated as

$$\varpi_{k,C_f} = \begin{cases} e^{-\lambda_1 T_{k,j}}, & \text{From } V_j \\ e^{-\lambda_2 T_{k,l}}, & \text{From } R_l \\ e^{-\lambda_3 T_{k,c}}, & \text{From } R_c \\ e^{-\lambda_4 T_{k,M}}, & \text{From MBS} \end{cases}, \tag{15}$$

where $\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 1$, and $0 < \lambda_1 < \lambda_2 < \lambda_3 \ll \lambda_4$.

To address the overestimation challenge in traditional DQN, we improve our cooperative caching strategy by utilizing DDQN consisting of two action value functions that estimate the action and its value [47]. In traditional DQN algorithm, the target Q value when the state is a non-terminal state at a certain moment is calculated as

$$y_t = r_t + \gamma \max_{a^*} Q(s_{t+1}, a^*; \theta^*), \tag{16}$$

DDQN does not directly select all possible Q values calculated by the target network by maximizing, but first selects the action corresponding to the maximum Q value by estimating network, and then target network calculates the target Q value according to the action,

$$y_t = r_t + \gamma Q(s_{t+1}, a_{\max}; \theta^*), \tag{17}$$

where $a_{\max} = argmax_a Q(s_{t+1}, a; \theta)$. Thus the target Q value of DDQN can be expressed as

$$y_t = r_t + \gamma Q(s_{t+1}, argmax_a Q(s_{t+1}, a; \theta); \theta^*), \tag{18}$$

where $\theta^* = \eta\theta^* + (1-\eta)\theta$, η represents the learning rate. The minimized loss function is given by

$$\omega(\theta) = \mathbb{E}[(y_t - Q(s_t, a_t; \theta))^2]. \tag{19}$$

The loss function can be optimized using Adam gradient descent,

$$\nabla_\theta \omega(\theta) = \mathbb{E}[(y_t - Q(s_t, a_t; \theta)\nabla_\theta Q(s_t, a_t; \theta))]. \tag{20}$$

After the scheduled iteration is completed, we can obtain a optimal cooperative caching scheme based on DDQN by combining the double time-scale content caching model with dynamic vehicle clustering. The specific process is shown in Algorithm 1.

Algorithm 1 DDQN-based cooperactive caching algorithm.

Initialization: Initialize action-value function $Q(s, a; \theta)$;
Initialize parameters of prediction and target network;
Initialize experience replay buffer D.
Input: Regional feature for each RSU;Vehicle clustering situation;Contents of vehicle request collected at period t_x
for each step of episode **do**
Observe the initial state s
for each time slot **do**
Select actions $\delta(t_x)$ and $\sigma(t_x^y)$ based on ε-greedy method;
Execute action $a(t_x^y)$, derive y_t according to (18) and obtain the next state $S(t_x^{y+1})$;
Store $(S(t), a(t_x^y), \varpi_{k,C_f}(t_x^y), S(t+1))$ into D;
Randomly get a batch of samples from D, and calculate target value y_t and update θ by minimizing loss function (19);
Optimize the loss function based on (20);
Update target networks as $\theta^* \leftarrow \eta\theta^* + (1-\eta)\theta$;
end for
end for

5. Simulation and result analysis

We constructed a simulation scenario of the edge caching network based on real traffic data set, which comprises historical location data for approximately 500 taxis in the San Francisco Bay Area over a month,

Table 2
Simulation Parameters.

Parameter	Value
Number of MBS	1
Number of RSUs	54
Number of vehicles	[100, 200]
Bandwidth	[540 KHz, 2000 KHz]
MBS transmission power	35 dBm
RSU transmission power	33 dBm
Vehicle transmission power	30 dBm
Vehicle communication range	300 m
Speed of vehicles	[10 m/s, 30 m/s]
Number of contents	[1000, 3000]
Each content size	1kB
Request interval	[0, 20]Seconds

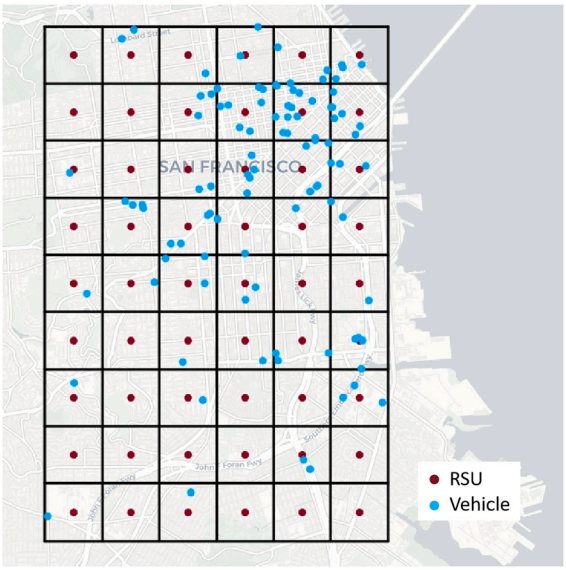


Fig. 7. Urban traffic grid division.

with an average interval of less than 10 seconds between successive location updates. We have delimited a 54 square kilometers (6 km × 9 km) region within the Bay Area and randomly selected 100 vehicles as participating vehicles, with all experiments performed in Python3.7 and tested on a processor (NVIDIA GeForce RTX 4060 Ti). Furthermore, we divided the Bay area into 54 traffic regions as shown in Fig. 7, each region representing a square area with a side length of 1 km and assigned an RSU. By analyzing the vehicle trajectories, we have identified that these areas may have varying road traffic conditions and traffic hubs, resulting in distinct regional feature attributes. The key simulation parameters are listed in Table 2.

In our constructed traffic scenario, the RSU will record the vehicle upon entering its coverage area. The traffic flow data collected by all RSUs in the past will be used as a dataset for predicting future traffic flow over a large time scale. For each prediction, 80% of the dataset will be randomly selected as training samples for the LSTM, while the remaining 20% will be used as test samples. Using LSTM, we predict the traffic flow for 54 RSUs, as shown in Fig. 8. The red inverted triangle curve represents the predicted traffic flow, while the black square curve represents the actual traffic flow.

In this paper, we evaluate the cache hit ratio, average transmission delay and communication overhead performance of our CRFDC in comparison with other schemes. Each result is obtained from the average of 20 simulations. With 100 vehicles, 50 cache capacity, 1000 content files, an average speed of 15 m/s, and a Zipf parameter of 1, we can report with 95% confidence that the cache hit ratio, transmission delay and communication overhead of our proposed scheme are [69.88%, 73.55%], [0.0311 ms, 0.0357 ms] and [2.67e-11 kWh, 2.96e-11 kWh],

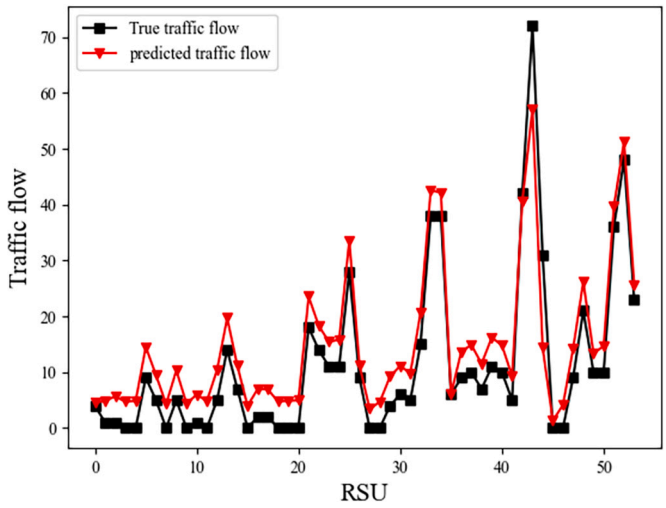


Fig. 8. Traffic flow prediction of RSUs.

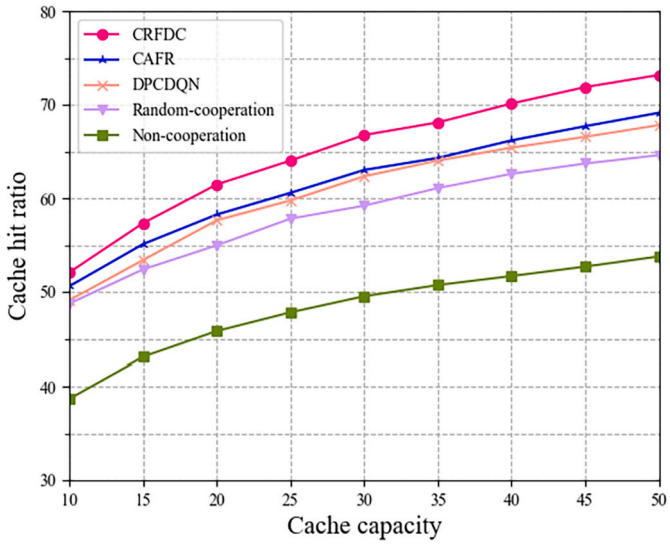


Fig. 9. Effect of RSU cache size on cache hit ratio.

respectively. We define cache hit ratio as the probability of successfully retrieving the required content in the edge cooperative nodes. It is considered a cache hit while vehicles receive the corresponding file from cooperative nodes through the V2V/V2R communication. In contrast, it is considered a cache miss if they receive the file from the MBS via V2M communication. We compare the performance of our CRFDC scheme with other existing researches and benchmark algorithms:

- CAFR: In the case of known content popularity, cooperative caching based on regional feature between RSUs can be utilized. The content is arranged in descending order of popularity, with the local RSU caching the first n popular items until the maximum cache capacity is reached. The remaining content is then cached by the cooperative RSU in a similar manner [48].
- DPCDQN: Considering a partitioned cooperative caching scheme that divides the cache buffers of RSUs into local static buffer and dynamic cooperative buffer, where local popular content is cached in one part and the remaining popular content in nearby regions is cached in the other part [49].
- Random-cooperation: In this approach, the RSU cooperative sets and vehicle cooperative nodes are randomly selected, and the decision of content placement/update process is executed based on DDQN algorithm.

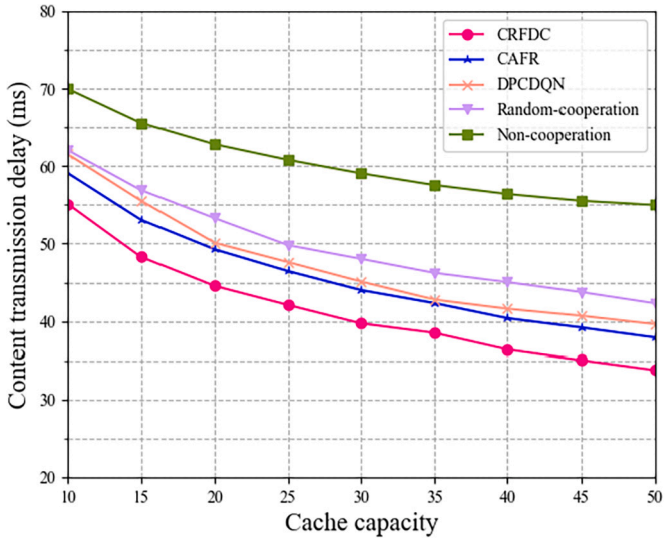


Fig. 10. Effect of RSU cache size on transmission delay.

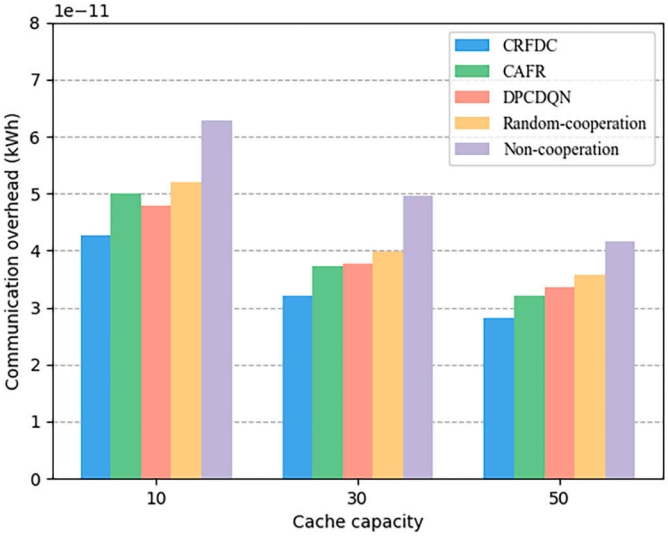


Fig. 11. Effect of RSU cache size on communication overhead.

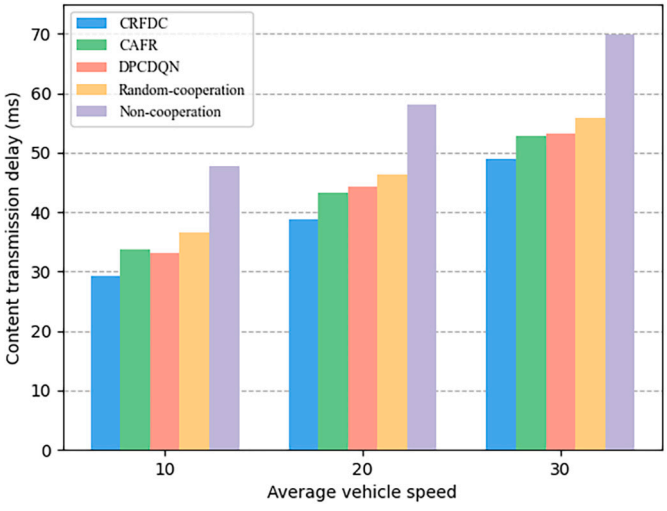


Fig. 12. Effect of vehicle speed on transmission delay.

- Non-cooperation: The RSUs and a randomly selected portion of vehicles independently cache popular content without considering the cooperation between RSUs or vehicle nodes.

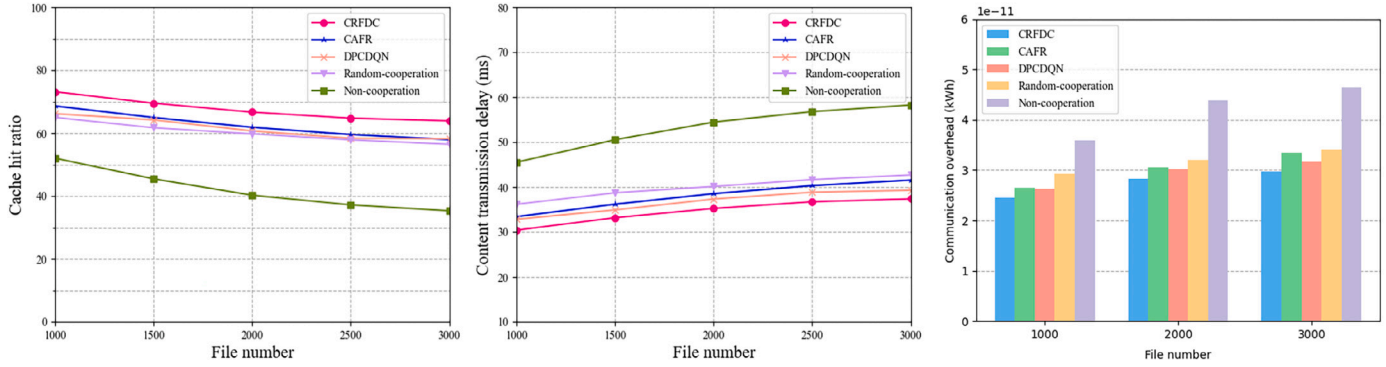
Fig. 9 show a comparison of the cache hit ratio for different schemes with varying RSU cache capacities. The average vehicle speed is 15 m/s, and there are 1000 content files. When initializing content popularity, we set $\alpha = 1$, which represents the exponent characterizing the Zipf distribution. The RSU cache capacity ranges from 10 to 50, increasing by 5 each time. Our CRFDC scheme shows the best performance, with a 4% and 5% improvement in the cache hit ratio compared to CAFR and DPCDQN, respectively. This can be explained by the fact that CRFDC uses DDQN to dynamically select the number of RSU cooperations based on traffic conditions and then utilizes consistent hashing to allocate content for caching nodes, which can effectively reduce cache redundancy between adjacent nodes. The cache hit ratio of CRFDC is 8% higher than the Random-cooperation scheme. This is because the Random-cooperation scheme does not consider regional feature when selecting RSU cooperations, instead randomly selecting them. This ignores the impact of traffic conditions on the cooperative caching performance and results in an unstable vehicle cluster where only randomly selected nodes are allowed to cache content, limiting its performance. The Non-cooperation scheme performs the worst among the five schemes because each cache node only focuses on its own cache condition and independently caches popular content. This leads to the waste of idle RSU cache resources and serious caching redundancy.

Fig. 10 illustrates the comparison of average content transmission delay among five caching schemes with varying RSU cache capacities. As shown in the curves, there is a downward trends as the RSU cache capacity increases. Our CRFDC outperforms CAFR, DPCDQN and benchmark algorithms due to its consideration of traffic conditions on RSU cooperative caching performance and its method of dividing vehicle clusters based on bridging centrality, which helps maintain stable connections between clusters. Additionally, CRFDC utilizes consistent hashing to allocate content for edge cooperation nodes, reducing cache redundancy and improving system stability. Furthermore, the content update framework based on DDQN ensures that CRFDC can adapt to the dynamic vehicle networking environment and further decrease content transmission delay.

The average communication overhead with different cache capacities of five schemes are depicted in Fig. 11. We define the communication overhead as the average amount of energy required per content request completed in a scheme. When cache capacity takes a small value, the content requests of vehicle users are mostly completed by MBS, and the high latency caused by long distance transmission of content will increase the communication overhead observably. With an increase in cache capacity, a greater number of popular contents can be cached in edge nodes, allowing the majority of users to obtain their requested content from nearby nodes, resulting in a significant reduction in communication overhead. When the cache capacity is 50, CRFDC reduces the communication overhead by the 12.1%, 16.3%, 21.4%, 32.3%, compared with CAFR, DPCDQN and two benchmark algorithms, respectively.

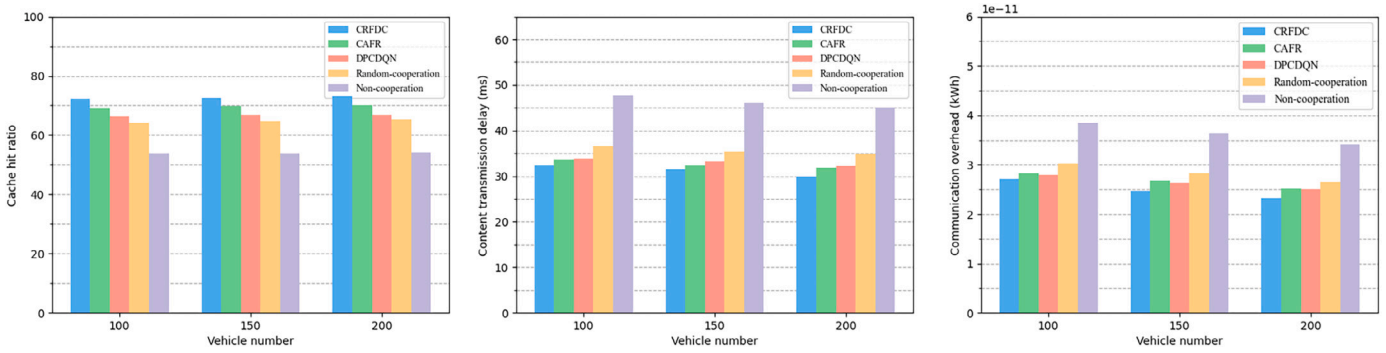
To evaluate the effect of vehicle mobility on transmission delay, we set the average vehicle speed to 10 m/s, 20 m/s and 30 m/s respectively. The average transmission delay increase with the increase of speed in Fig. 12, it can be explained that as the vehicle speed increases, the stability of vehicle cluster link becomes worse, and the vehicle communication condition drops sharply. The performance of CRFDC is consistently better than the other four schemes, as evidenced by the fact that when the average speed is 30 m/s, the transmission delay of CRFDC is approximately 7% lower than CAFR, about 8% lower than DPCDQN and approximately 14% and 30% lower than Random-cooperation scheme and Non-cooperation scheme, respectively.

In this set of simulations, we gradually increase the number of content files from 1000 to 3000 and evaluate its impact on cache hit ratio,



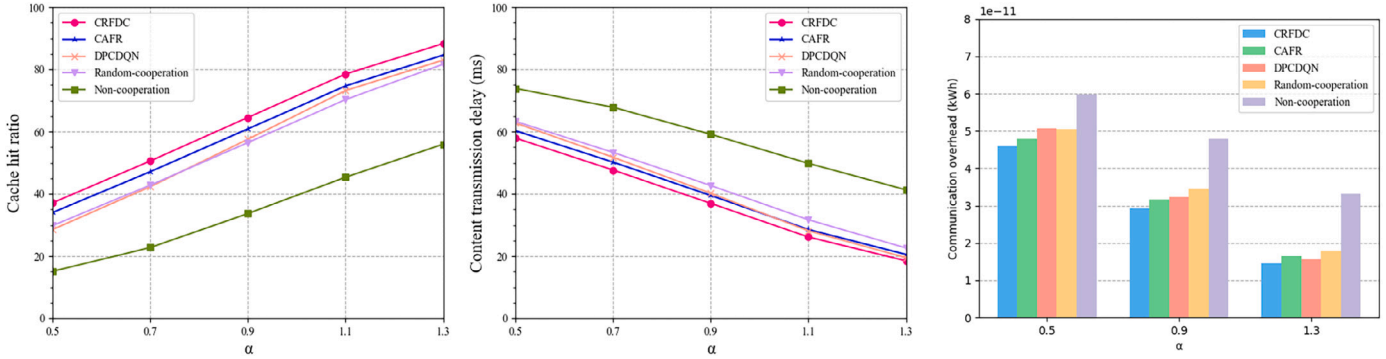
(a) cache hit ratio (b) transmission delay (c) communication overhead

Fig. 13. Effect of file number.



(a) cache hit ratio (b) transmission delay (c) communication overhead

Fig. 14. Effect of vehicle density.



(a) cache hit ratio (b) transmission delay (c) communication overhead

Fig. 15. Effect of Zipf parameter.

transmission delay and communication overhead. In Fig. 13a, the cache hit ratios show a downward trend as the number of content files increases. This decrease can be attributed to a higher demand for content requests from users. Due to the limited cache capacity of the cache node, it is unable to meet the diverse requirements of content requests, resulting in more users having to obtain files from remote MBS. Therefore, the average transmission delay shows an upward trend in Fig. 13b. A wider range of content makes it more difficult for vehicles to obtain content from edge caching nodes, increasing the communication overhead in Fig. 13c. All the while our CRFDC outperforms the other four schemes since it takes the RSU cooperative caching based on regional feature, dynamic vehicle clustering and consistent hashing allocation into consideration.

Fig. 14 shows the performances versus vehicle density for different strategies while the quantity of content files is 1000, and the RSU cache size is 50. In Fig. 14a, when the number of vehicles gradually increases, the cache hit ratios of five schemes remain almost unchanged, it can be explained that as more vehicles enter the RSU coverage area, the increase in the number of user requests is much higher than the increased cache capacity of the vehicle as an edge caching node. In Fig. 14b, the content transmission delay decreases slowly as the number of vehicles increases. This is because denser vehicles means closer communication distance between vehicles in a vehicle cluster, making the content communicated via V2V transmits faster and in less time. In a similar vein, most users can obtain requested content in V2V mode, leading to less communication overhead in Fig. 14c.

In Fig. 15, the Zipf parameter varies from 0.5 to 1.3. Fig. 15a shows the cache hit ratio increases with α. A larger α indicates the more concentrated content distribution, which means it is easier for edge nodes to cache the more popular content, resulting in a higher cache hit ratio. In Fig. 15b, the transmission delay curves show a downward trends as α increases. This is because the highly concentrated content distribution

allows most content requests to be completed by edge nodes, eliminating the need for users to obtain the requested content from remote MBS through backhaul links with high latency, which saves a significant amount of transmission time. Similarly, more RSU can cache the most popular content and vehicles can obtain more contents from edge caching nodes, thus reducing the communication overhead in Fig. 15c.

6. Conclusion

In this paper, we have investigated the issue of double time-scale edge caching in vehicular social networks. Based on real traffic data, we have developed a cooperative caching model. Our proposed edge cooperative caching scheme utilizes regional feature and dynamic vehicle clustering to facilitate flexible cooperation between RSUs and intelligent vehicles. This allows for efficient utilization of the storage resources available at edge nodes, while also adapting to the dynamic vehicle networking environment. The simulation results demonstrate that our scheme can significantly improve the cache hit ratio, reduce the average transmission delay and decrease the communication overhead for vehicle users to access content.

We are continuously working to enhance our proposed caching scheme. To effectively handle the dynamics and complexity of the vehicle communication environment, we are focused on improving the learning ability of our cache system. Additionally, we plan to further optimize our caching scheme by considering the cooperative management of vehicular social characteristics management and resource allocation.

CRediT authorship contribution statement

Yujian Chen: Writing – original draft, Validation, Software, Resources, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Weidi Tian:** Software, Project administration, Methodology, Investigation. **Zhengle Li:** Data curation, Formal analysis, Software. **Hui Song:** Writing – review & editing, Visualization, Validation, Supervision, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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